- Oracles
- Deutsch Algorithm:

- input classical data to algos
- encode classical functions in circuits

$$\begin{cases} \vec{b} = (b_{n-1}, b_{n-2} \dots b_2, b_1, b_0) \\ f(\vec{b}) = \vec{c} \end{cases}$$

$\vec{b}$	$\vec{c}$
000	001
001	010
010	011
011	100
100	101
101	110
110	111
111	000

$F(k) = (k+1) \bmod 8.$

$$\vec{f} = \underline{|001\rangle}\,\underline{\langle 000|} + |010\rangle\langle 001| + \dots + ?$$

problem: if $\vec{f}$ not invertible

$$O_f|\vec{b}\rangle = |\vec{c}\rangle$$
$$= |\vec{f}(\vec{b})\rangle$$

if there are 2 inputs $\vec{I}$ that are diff (orthogonal) but produces the same output $\vec{c}$, no unitary can do that.

↳ unitaries will produce orthogonal outputs

$$\Updownarrow$$

$$\vec{f}(\vec{x}) = \vec{f}(\vec{y}) \text{ for } \vec{x} \neq \vec{y}$$

$$O_f|\vec{x}\rangle = O_f|\vec{y}\rangle \rightarrow$$

if I have 3 input bits → 1 output bit

$$\Downarrow$$

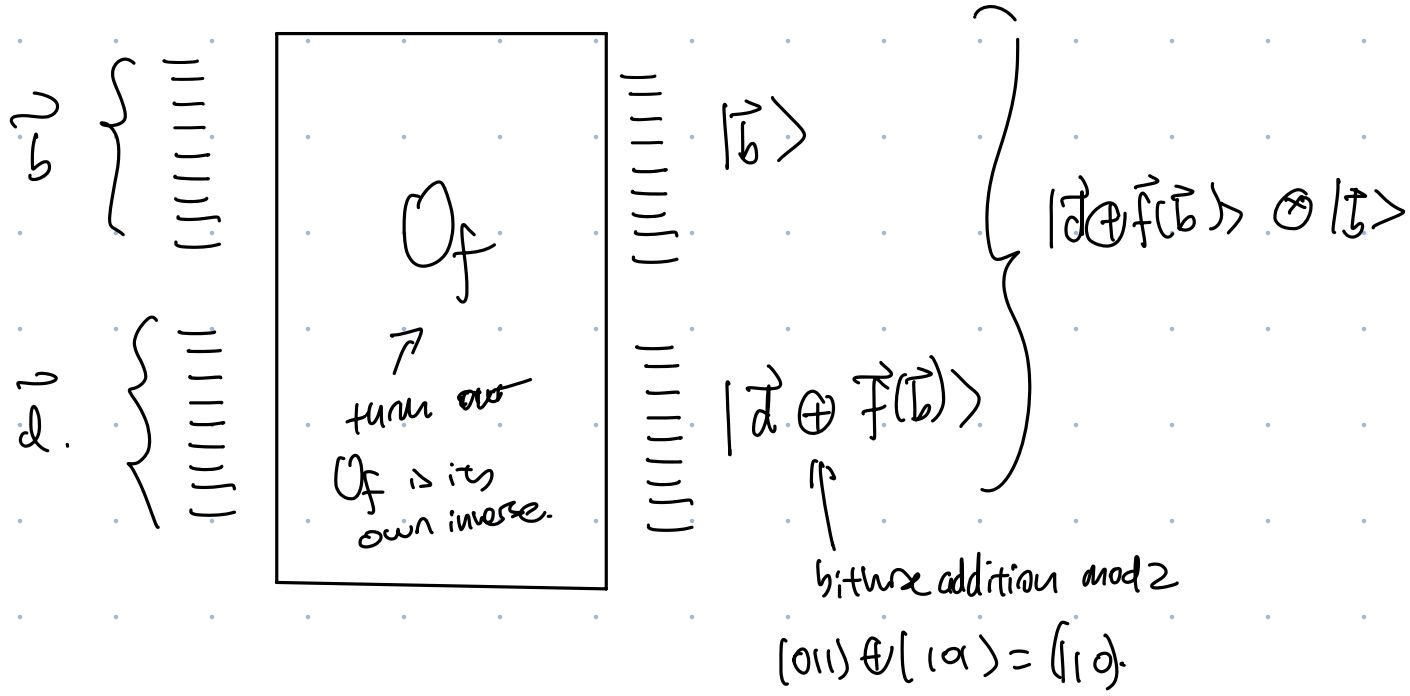
not invertible!

↳ $\langle \vec{y} | \underbrace{O_f^\dagger O_f}_{I} | \vec{x} \rangle = 1$ } contradiction.

$= 0$

We are now going to generalize the trick we learned in Toffoli:

↳ Solution:



$$\vec{O}_f^{\,2}\left(|\vec{b}\rangle \otimes |\vec{d}\rangle\right) = |\vec{d} \oplus \underbrace{\vec{f}(\vec{b}) \oplus \vec{f}(\vec{b})}_{0}\rangle \otimes |\vec{b}\rangle$$

$$= |\vec{d}\rangle \otimes |\vec{b}\rangle$$

Hence $\vec{O}_f^{\,2} = I$.

more justification?

Hermitian: real symmetric

Unitary

Hence oracles are legal!

$|\vec{d} + \vec{f}(\vec{x})\rangle \otimes |\vec{x}\rangle$ and $|\vec{d} + \vec{f}(\vec{y})\rangle \otimes |\vec{y}\rangle$ are still orthogonal!

Now, even if $|\vec{d} + \vec{f}(\vec{x})\rangle = |\vec{d} + \vec{f}(\vec{y})\rangle$

Deutsch Algorithm 1985.

↳ Did smt completely impossible classically.

Alice creates an oracle O_f for a binary func.

$\{0,1\} \to \{0,1\}$ ($N=1$). (how many func.)

$(2^M)^{(2^n)}$

$\Downarrow$

only 4 choices!

balanced { Identity, NOT

constant { ZERO, ONE

Alice gives Bob access to the blackbox oracle.

Bob's task is not to find which function is encoded (need 2 bits info)

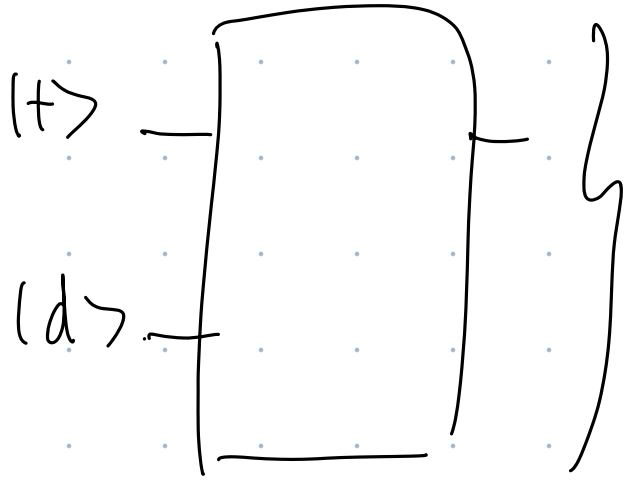
↳ Bob's task is to answer:

↳ f constant or balanced?

Task is impossible in 1 guess.

ex. Suppose $f(0)=1$, f would be ONE or NOT.

Needs a second guess $f(1)$



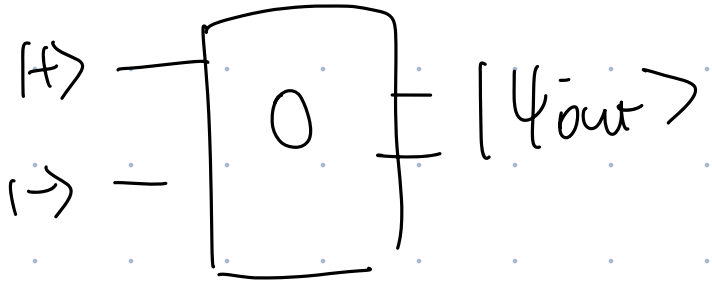
$$\frac{1}{\sqrt{2}}\left\{ |\vec{d}+\vec{f}(0)\rangle \otimes |0\rangle + |\vec{d}+f(\vec{1})\rangle \otimes |1\rangle \right\}$$

↳ How does it distinguish

Need to do more:

$$g = f(0) \oplus f(1)$$

$$\boxed{\begin{array}{l} g=0 \Rightarrow \text{constant} \\ g=1 \Rightarrow \text{balanced.} \end{array}}$$



$$|\psi_{in}\rangle = |-\rangle \otimes |+\rangle = \frac{1}{2}\{ |0\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle \}$$

$$|\psi_{out}\rangle = \frac{1}{2}\{ |0 \oplus f(0)\rangle \otimes |0\rangle + |0 \oplus f(1)\rangle \otimes |1\rangle - |1 \oplus f(0)\rangle \otimes |0\rangle - |1 \oplus f(1)\rangle \otimes |1\rangle \}$$

If f is constant:

$$f(0) = f(1)$$

$$|\psi_{out}\rangle = \frac{1}{2}\Big\{ |0 \oplus f(0)\rangle \otimes |0\rangle + |0 \oplus f(0)\rangle \otimes |1\rangle - |1 \oplus f(0)\rangle \otimes |0\rangle - |1 \oplus f(0)\rangle \otimes |1\rangle \Big\}$$

$$= \frac{1}{2}\Big\{ |0 + f(0)\rangle \otimes (|0\rangle + |1\rangle) - |1 \oplus f(0)\rangle \otimes (|0\rangle + |1\rangle) \Big\}$$

$$= (-1)^{f(0)} \; |-\rangle \otimes |+\rangle$$ → measure in th

info I don't want to learn hidden in global phase!

get the same state.

If f is balanced, $f(0) \oplus f(1) = 1$.

$\hookrightarrow$ $0 \oplus f(0) = 1 \oplus f(1)$, $0 + f(1) = 1 \oplus f(0)$.

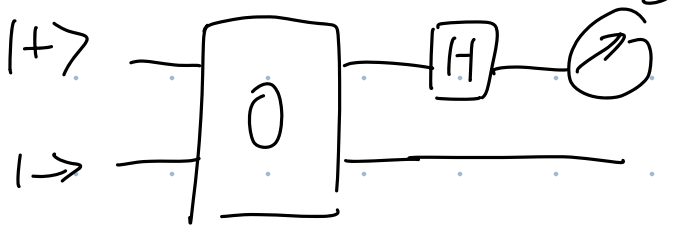
$$|\psi_{out}\rangle = \frac{1}{\sqrt{2}}\Big\{ |0 + f(0)\rangle \otimes |0\rangle + |0 \oplus f(1)\rangle \otimes |1\rangle - |0 \oplus f(1)\rangle \otimes |0\rangle - |0 + f(0)\rangle \otimes |1\rangle \Big\}$$

$$= \frac{1}{2}\Big\{ |0 + f(0)\rangle \otimes (|0\rangle - |1\rangle) + |0 \oplus f(1)\rangle \otimes (|1\rangle - |0\rangle) \Big\}$$

$= \pm\left(|-\rangle \otimes |-\rangle\right)$

$H|+\rangle = |0\rangle$

$H|-\rangle = -|1\rangle$?



constant f : measure +1.

balanced f : measure −1.

this is very cool!!!